

Contribution: 30%

Brief:

You are required to implement pytest unit testing on all the functions in the provided app.py file from the git repository below.

You are permitted to use your own notes however you are **not** permitted to access the Internet during this practical.

You **may not use** generative artificial intelligence (AI) (ChatGPT, CoPilot, ChatSonic, Bing Chat, Lex, DALL-E 2, or other similar tools) Suggest you turn off co-pilot in VSC for this,

Notes:

- 1) The purpose of this practical is to design and write the tests per the function specification in this document **DO NOT** correct any code errors or test per the code.
- 2) You are required to follow the good practice of using a separate test folder and also of providing only 1 test assert per unit test and use Equivalence Partition and Boundary Value Analysis to determine the inputs to the tests (detail in an excel spreadsheet if desired).
- 3) A function may require more than one test. The use of comments for the assert is recommended.
- 4) The use of parametrization and test fixtures where appropriate is expected.

Setup

Clone the project folder using

Git clone https://github.com/goslingp/groupB_Practical1

This should produce a folder named sys_test_ca1 which contains the file to be tested – **pytest_pract1.py**

Submission:

You may submit your work (artifacts to include the project folder, excel workbook, document of results to Moodle as either

- A) A text file with the link to your git repository with peter.gosling@dkit.ie as a team member.

or

- B) A zip file of your project folder, excel and results document,

Upload to Moodle **BEFORE** leaving the lab (do not make changes to your git repository after the class)

Function details and marking scheme

Feature	Function	Max Marks
Use of a tests folder	Correct test folder setup and use	5
Excel workbook	Outline the inputs and expected values for tests	5
fahrenheit_to_celsius()	This function should convert a temperature float of Fahrenheit degrees to a float of Celsius degrees Input Selection: 5 Parametrise: 3 Correct: 2	10
reverse_string()	Function to return the reverse of an entered string Input Selection: 5 Parametrise: 3 Correct: 2	10
add()	Performs addition on two integer numbers, returns an integer number. Input Selection: 5 Parametrise: 3 Correct: 2	10
classify_student_record()	Returns student name : classification of a students score in a record of name, mark (0-100). Marks ≥ 70 achieve First Class, 60 to 69 is 2.1, 50 – 59 is 2.2, 40 to 49 is Pass and below 40 is Fail . Error messages: Invalid record format for incorrect input formats, Score must be a number for non numeric, Score out of range marks <0 or >100 . Input Selection: 5 Parametrise: 3 Correct: 2	10
calculate_team_points()	Calculate the number of points a team has based on results read from a text file – results.txt is provided. Returns number of points achieved. Rules: a) 3 points for a win b) 1 point for a draw c) 0 point for a game lost Input Selection: 5 Parametrise: 3 Fixture: 3 Correct: 2	13
Include a coverage report for the project	Marks provided on total coverage	9
Include a html report	Marks provided for html report	8
Include the commands / results	Record the commands you used to run the tests and screenshots of the results in a word or similar document	10
Upload complete project code	Upload ALL your files for this practical either via Moodle (zip file) A git repository with peter.gosling@dkit.ie as a team member.	5 Or 10
Total		100

Coverage report -m to get report